Rain Model for Events

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1 Climatology

1.1 Estimation of the probability of rain

We start from the fact that rain exceeding a certain threshold t (for example 2 mm) on a given day D follows a Bernoulli distribution:

$$X_D \sim \text{Bernoulli}(p) \tag{1}$$

If we consider that the X_D are i.i.d., the number of rainy days in a given time interval follows a binomial distribution:

$$K = \sum_{i=1}^{n} X_i, \qquad K \sim \text{Bin}(n, p), \qquad \Pr(K = k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{2}$$

The interval we consider contains all the days within ±7 days of the event day, over the past 30 years. This means that $n \approx 450$, and k is the number of days with rain exceeding our threshold t.

We know from intuition, and directly from the maximum likelihood estimator (MLE), that the best estimator of p is

$$\hat{p} = \frac{k}{n} \tag{3}$$

with k the number of rainy days exceeding t and n the total number of days in our sample.

Considering such an interval first allows us to use much more data than only the exact same day of the previous years. Second, it gives us a large enough n to use the Central Limit Theorem to compute confidence intervals.

1.2 Moments

From the binomial distribution (2), by linearity of the expectation and independence of the X_i, we know that:

$$\mathbb{E}[K] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np \qquad \text{and} \qquad \text{Var}(K) = \sum_{i=1}^{n} \text{Var}(X_i) = np(1-p) \tag{4}$$

Therefore, with $\hat{p} = K/n$ from (3), we compute the expected value and the variance of $\hat{p}$ using (4):

$$\mathbb{E}[\hat{p}] = \frac{1}{n}\,\mathbb{E}[K] = \frac{np}{n} = p \qquad \text{and} \qquad \text{Var}(\hat{p}) = \frac{1}{n^2}\,\text{Var}(K) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n} \tag{5}$$

1.3 De Moivre–Laplace theorem and CLT

We can approximate the binomial distribution by a normal distribution if these two conditions are met:

- $np \geq 10$
- $n(1-p) \geq 10$

With $n = 450$, these two conditions are verified if $p \in [2.2\,\%,\ 97.8\,\%]$.

Then, using the moments (4) and (5), we know that:

$$K \ \approx\ \mathcal{N}\big(np,\ np(1-p)\big) \qquad \text{and} \qquad \hat{p} = \frac{K}{n} \ \approx\ \mathcal{N}\left(p,\ \frac{p(1-p)}{n}\right) \tag{6}$$

This result gives the approximate distribution of our estimator $\hat{p}$, which we can standardize to build tests and confidence intervals more easily:

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \ \approx\ \mathcal{N}(0,1) \tag{7}$$

1.4 Confidence interval and inference

From (7), for a confidence level $1-\alpha$ we have:

$$\Pr\left(|\hat{p} - p| \leq z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) \approx 1-\alpha \tag{8}$$

with $z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$.

The Wald confidence interval replaces p by $\hat{p}$ in the square root of (8), so that p can be isolated directly. This leads to the interval:

$$\mathrm{IC}_{1-\alpha}(p) = \left[\hat{p} - z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\ ,\ \ \hat{p} + z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right] \tag{9}$$

The problem of the Wald confidence interval is that it comes from two approximations. First, the CLT, which only works under the two conditions we saw (it does not work for p close to 0 or 1). Second, the use of $\hat{p}$ instead of p in the standard error. Because of this second approximation, the interval (9) can go below 0 or above 1, and collapses to $[0,0]$ when $k = 0$.

To compensate for this second problem, we preferred the Wilson interval.

1.5 Wilson interval

The Wilson interval still relies on the CLT, but it removes the second approximation: it keeps p in the standard error. For each candidate value p_0, we test $H_0 : p = p_0$ using (8), and the interval contains all the values of p_0 that are not rejected:

$$\left\{ p_0 \in [0,1]\ :\ |\hat{p} - p_0| \leq z\sqrt{\frac{p_0(1-p_0)}{n}} \right\}, \qquad z = z_{1-\alpha/2} \tag{10}$$

Squaring the inequality in (10) gives a second-degree polynomial in p_0, whose two roots are the bounds of the interval:

$$\boxed{p_\pm = \frac{\hat{p} + \dfrac{z^2}{2n} \ \pm\ z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}} \tag{11}$$

2 Forecast

The problem with the climatology inference is that it only considers past events and **not** the current state of the atmosphere and the current weather. We therefore need to include the forecast, which is much more precise if the event is soon but loses its value after roughly 2 weeks. The main idea here is to trust climatology more if the event is far away, and to put more emphasis on the forecast if the event is close to the current date.

2.1 Weight function

We therefore consider a weight function $w(h)$, with h the number of days between today and the event, and define our final estimate as follows:

$$p = w(h)\, p_f + \big(1 - w(h)\big)\, \hat{p}, \qquad w(h) = \max\!\left(0,\ 1 - \frac{h}{16}\right) \tag{12}$$

with p_f the probability given by the forecast and $\hat{p}$ the probability given by the climatology (3). The number 16 comes from the fact that the API we use stops making forecasts after 16 days; it is not a statistical choice.

As we can see, we defined a linear weight for now, which is a possible improvement for the future. A good method could be to use the Brier score, choosing for each horizon h the weight that minimises the quadratic error between the given probability and what really happened.

Another limitation of the model is that p_f represents the probability of any rain (even a rain of 0.1 mm), whereas $\hat{p}$ represents the probability of rain exceeding the threshold t. As we want to interpret p in (12) the same way as $\hat{p}$, we tend to overestimate p, because p_f drives p higher than it should be. A correction using conditional probability could be a good idea.

2.2 Confidence interval

We decide to assume that p_f is deterministic and has no variance (meaning that the probability given by the API is the real probability). This is a strong assumption and is not expected to be true, but we make it for the sake of simplicity. Applying the weights of (12) to the Wilson bounds (11), we compute the confidence interval as:

$$[\, p_{\text{lo}},\ p_{\text{hi}}\,] = w\, p_f + (1 - w)\,[\, p_-,\ p_+\,] \tag{13}$$

where $w = w(h)$ is the weight for the given horizon h.

3 Decision

3.1 Loss without protection

On the event day D, the rain is $X_D \sim \text{Bernoulli}(p)$, with p given by (12), and the organizer's loss L is:

$$L = \lambda R \cdot X_D = \begin{cases} \lambda R & \text{with probability } p, \\ 0 & \text{with probability } 1 - p, \end{cases} \tag{14}$$

where R is the total revenue of the event and λ the share of the revenue lost in case of rain above the threshold t.

It means that the expected value of the loss is:

$$\mathbb{E}[L] = \lambda R\, p \tag{15}$$

3.2 Loss with protection

We suppose here that the protection is perfect, meaning that for a certain cost C, paid with certainty, the loss is brought back to 0.

3.3 Decision threshold

If the organizer compares the expected losses, they protect themselves when the protection is cheaper than the expected loss (15). In other words, when:

$$C < p\,\lambda R \quad \Longleftrightarrow \quad p > \frac{C}{\lambda R} \tag{16}$$

The threshold is then defined by:

$$\boxed{p^* = \frac{C}{\lambda R}} \tag{17}$$

which is a classic cost–loss model, with p^* being the cost–loss ratio.

3.4 Interpreting p^*

p^* is the price of the protection as a percentage of what we lose in case of a rainy day. The rule (16) is therefore pretty easy to understand: "protect yourself if the probability of rain exceeds the price of the protection divided by the potential loss."

Here an important assumption is made regarding the organizer: **risk neutrality**. This assumption is often not realistic, in particular for small organizers. Indeed, a potential loss of 12 000 CHF with a probability of 25 % may not be judged the same way as a certain loss of 3 000 CHF, because the 12 000 CHF loss could be too damaging. This could be improved by modelling the risk aversion of the organizer, or by adding a constraint on the maximal accepted loss.

3.5 Dealing with the uncertainty of p

If p were known exactly, there would be only two cases: $p > p^*$ or $p < p^*$. But p is estimated, with the interval $[p_{\text{lo}}, p_{\text{hi}}]$ from (13). Depending on where the threshold p^* (17) lies relative to this interval, the verdict is different:

$$\text{verdict} = \begin{cases} \textbf{Protect} & \text{if } p_{\text{lo}} > p^* \quad \text{(the whole interval is above the threshold)} \\ \textbf{Don't protect} & \text{if } p_{\text{hi}} < p^* \quad \text{(the whole interval is below the threshold)} \\ \textbf{Close call} & \text{if } p_{\text{lo}} \leq p^* \leq p_{\text{hi}} \quad \text{(the threshold is inside the interval)} \end{cases} \tag{18}$$

3.6 Other assumptions

Other assumptions have been made for the sake of simplicity, and they are all possible ways to improve our model. Statistics is not about finding the truth, but about getting as close as possible to it :)

Assumption	**Reality**
Binary rain (above or below t)	The loss depends on the intensity and duration of the rain.
Fixed λ	The loss also depends on when the rain falls (in the morning or during the concert).
Perfect protection	A tent does not save everything: some residual loss usually remains.
Single day	A three-day festival requires combining several days, which are correlated with each other.

Table 1: Assumptions of the decision rule